
Audio-Centric Visual Speech Reconstruction with Weak Text Conditioning

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TL;DR

This report summarizes the final audio generation method developed in the FM-AVSR branch. The key finding is that high-quality visual speech audio generation should not be treated as a strict vision-to-text-to-audio cascade. In our final system, text is only a weak semantic auxiliary condition; the main predictive path is from lip motion and audio-derived latent conditions to continuous Mimi audio latents. This design avoids making transcript accuracy the bottleneck. Replacing ground-truth text with Auto-AVSR text changes full-validation latent correlation from 0.5818 to 0.5783, a drop of only 0.0035, while explicit timbre/audio prompt conditioning and residual Mimi-latent reconstruction provide the major gains. The final model uses Auto-AVSR visual features, SmolLM2 text hidden states, speaker features, a 3.04-second Mimi audio prompt, and prompt-level timbre statistics to reconstruct normalized Mimi latents. The strongest checkpoint reaches 0.58184180 mean correlation on the LRS3 validation split and supports raw-video and silent-video inference with optional reference audio.

1 Goal and Main Claim

The project goal is to generate speech audio from talking-face video. The final system is an offline research prototype: it predicts Mimi audio latents from visual speech conditions and decodes them into waveform. It is not positioned as a real-time product system or a full speaker-cloning model.

The central conclusion is architectural. A direct *vision* $\rightarrow$ *text* $\rightarrow$ *audio* pipeline is too dependent on transcript quality and discards acoustic information that is visible in lip dynamics but not represented by text. Our final route instead uses:

$$\text{lip/video} + \text{weak text} + \text{speaker/timbre} \longrightarrow \text{Mimi audio latent} \longrightarrow \text{waveform}. \quad (1)$$

Text helps stabilize content, but it is not the main representation used to generate audio. This explains why imperfect AVSR text can still produce reasonable audio: the model is not forced to reconstruct speech from text alone.

The final technical contributions are:

1. **Audio-centric latent formulation.** We formulate visual speech generation as reconstruction of normalized continuous Mimi latents rather than as TTS from a transcript. This preserves acoustic information beyond text and gives a direct objective for waveform reconstruction.
2. **Residual endpoint reconstruction architecture.** The final model abandons stochastic FM sampling for a deterministic endpoint predictor with a frozen base reconstructor and a learned residual correction.
3. **Explicit timbre/audio prompt conditioning.** The first 38 Mimi frames, about 3.04 seconds, are used both as prompt tokens and as global mean/std timbre statistics. This provides speaker and recording-color cues that text cannot provide.

4. **Weak text conditioning finding.** Ground-truth text and Auto-AVSR text give nearly identical latent correlation. This shows that, in this architecture, exact transcript accuracy is not the dominant bottleneck; the main load is carried by lip and audio-latent conditions.
5. **Data processing as model evidence.** The final results depend on a consistent data representation: AVSR-compatible lip crops, Auto-AVSR visual latents/text, SmolLM2 hidden states, Mimi targets, speaker features, and prompt-derived timbre conditions.

2 Data Processing and Data Use

Data construction is a core part of the method. Each clip is converted into a matched set of visual, text, speaker, and audio-latent files. The model never uses raw text alone as the generation target; all training and evaluation are defined against Mimi continuous latents.

2.1 Training and Validation Splits

The final checkpoint is trained on the lip-AVSR processed pretrain split with 59,144 training clips and evaluated on a fixed 1,000-clip validation split:

- train split: `configs/eval_splits/pretrain_lipavsr_train59144_seed44_plus_ready_remaining9144_excl_val1000_seed43.txt`
- validation split: `configs/eval_splits/pretrain_len80_260_lipavsr_val1000_seed43.txt`

Earlier 10k and 30k subsets were used to validate scale trends and ablations before moving to the final 59k setting.

2.2 Per-Clip Files

File	Shape	Use in the model
<code>lip_avsr.npy</code>	$(T_v, 96, 96)$	AVSR-compatible grayscale mouth crop
<code>avsr_enc_lipavsr.npy</code>	$(T_v, 768)$	visual/lip condition
<code>avsr_text_lipavsr.txt</code>	text	Auto-AVSR transcript for raw inference
<code>smollm2_h_text_json.npy</code>	$(L, 960)$	GT-text hidden states for in-domain eval
<code>smollm2_h_lipavsr.npy</code>	$(L, 960)$	AVSR-text hidden states for raw inference
<code>speaker_emb.npy</code>	(256)	face-based identity condition
<code>latent.npz</code>	$(T_a, 512)$	Mimi latent training target
<code>audio_prompt.npy</code>	$(38, 512)$	Mimi prompt tokens, first 3.04 s
<code>timbre_cond.npy</code>	(1024)	prompt mean/std timbre condition

Table 1: Final per-clip representation. The model consumes conditions and predicts the normalized Mimi latent target.

2.3 Visual Processing

The visual path must use the Auto-AVSR-compatible mouth crop `lip_avsr.npy`. The older RGB lip crop is not the final representation. The correct crop is encoded by Auto-AVSR:

$$e_{1:T_v}^v = \text{AutoAVSR}_{\text{enc}}(V_{\text{lip}}), \quad e_t^v \in \mathbb{R}^{768}. \quad (2)$$

Auto-AVSR also produces a decoded text string. For external videos without ground-truth text or timestamps, this decoded text becomes the text source.

This processing choice is important for generalization. On external raw videos, switching from the old RGB lip path to the correct `lip_avsr` path improved latent correlation:

Video	Old RGB lip path	Final lip-AVSR path
Trump	0.30353260	0.38691377
HRX	0.25592437	0.27240886

Table 2: External-video checks showing why the final visual representation matters. This is reported as data processing evidence, not as a standalone contribution.

2.4 Audio Latent Target

Mimi provides the continuous audio latent target. For each audio-latent frame $z_t \in \mathbb{R}^{512}$, the training target is normalized per latent dimension:

$$y_t = \frac{z_t - \mu}{\sigma}. \quad (3)$$

The model predicts $\hat{y}_{1:T}$, then denormalizes and decodes:

$$\hat{z}_t = \hat{y}_t \sigma + \mu, \quad \hat{x}_{\text{wav}} = \text{MimiDecoder}(\hat{z}_{1:T}). \quad (4)$$

All reported MSE, MAE, and correlation metrics are computed in normalized Mimi latent space, not waveform space.

2.5 Text and Timbre Conditions

Text is encoded by SmolLM2 into hidden states. For LRS3 validation with ground-truth word timestamps, audio-latent frames are aligned to text positions using word timestamps. For raw videos, timestamps are unavailable, so the hidden states are uniformly resampled. This makes text usable even when it is not perfectly recognized or perfectly aligned.

The timbre condition is derived from the first 38 normalized Mimi frames $A \in \mathbb{R}^{38 \times 512}$:

$$q = [\text{mean}(A), \text{std}(A)] \in \mathbb{R}^{1024}. \quad (5)$$

The same A is also provided as prompt tokens. Listening examples remove this 3.04-second prompt region when the prompt comes from the same clip, because the first seconds are condition input rather than generated speech.

3 Architecture and Mathematical Formulation

3.1 Condition Fusion

Figure 1 shows the final inference graph. The model condition is:

$$C = \{e_{1:T_v}^v, h_{1:L}, s, A, q\}, \quad (6)$$

where e^v is the Auto-AVSR visual sequence, h is the SmolLM2 text hidden sequence, s is the speaker embedding, A is the audio prompt token sequence, and q is the prompt statistics timbre vector.

Auto-AVSR visual features are produced at about 25 Hz, while Mimi latents are used at 12.5 Hz. The visual condition is aligned by taking every second visual frame:

$$v_t = e_{2t}^v. \quad (7)$$

3.2 Deterministic Residual Reconstruction

The explored flow-matching route learned a vector field from noise to target. It did not become competitive: sampling correlation stayed around 0.35, far below the reconstruction route. The final model therefore fixes the inference state to:

$$\tilde{x} = 0, \quad \tau = 1, \quad (8)$$

and directly predicts the endpoint latent. There is no random noise input and no iterative denoising loop.

The final checkpoint uses a frozen base reconstructor and a residual head:

$$y_{\text{base}} = f_{\text{base}}(C), \quad \Delta = f_{\text{residual}}(C), \quad \hat{y} = y_{\text{base}} + \Delta. \quad (9)$$

This residual design is the main architecture-level improvement because it lets the final model refine a strong endpoint predictor instead of relearning the whole mapping from scratch.

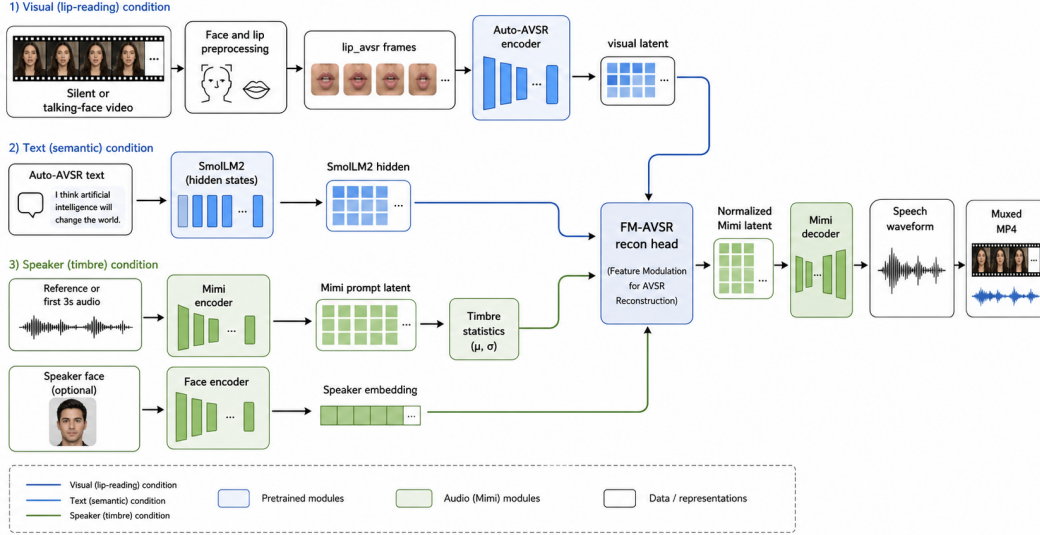


Figure 1: Final audio-centric visual speech reconstruction architecture. Lip features, weak text hidden states, speaker identity, and audio-prompt timbre conditions are fused to predict normalized Mimi latents.

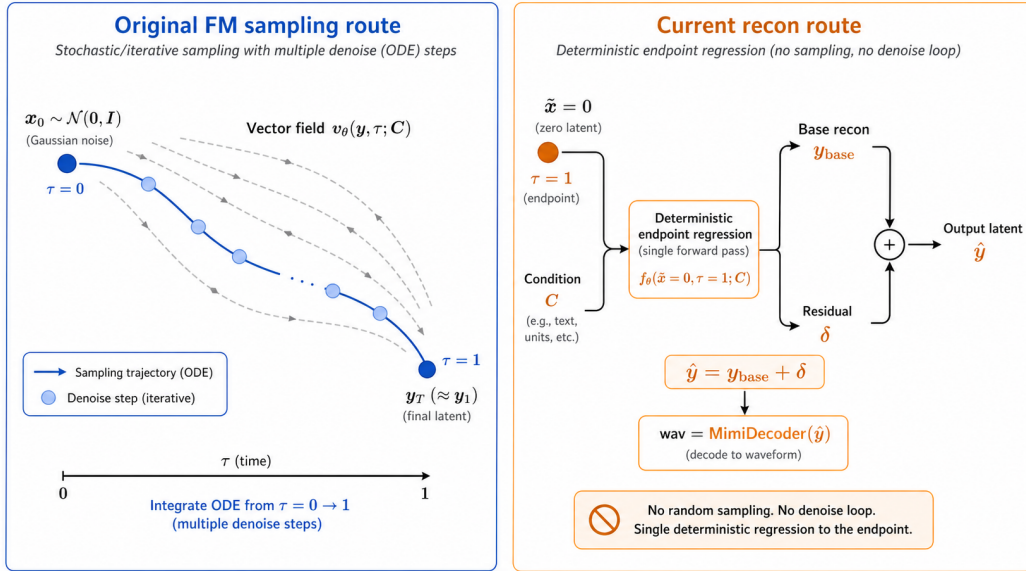


Figure 2: The final route is deterministic endpoint reconstruction, not stochastic FM sampling. The model fixes $\tilde{x} = 0$ and $\tau = 1$, then predicts Mimi latents through a frozen base plus residual correction.

3.3 Training Objective

The final loss combines reconstruction, sample-level correlation, and prompt timbre statistics:

$$\mathcal{L} = \mathcal{L}_{\text{mse}} + 0.2 \mathcal{L}_{\text{corr}} + 0.05 \mathcal{L}_{\text{prompt_stats}}. \quad (10)$$

The correlation term is computed per clip after skipping the first 38 prompt frames:

$$\mathcal{L}_{\text{corr}} = 1 - \text{corr}(\text{vec}(\hat{y}_{39:T}), \text{vec}(y_{39:T})). \quad (11)$$

The prompt-statistics term encourages the generated segment to stay close to the timbre statistics implied by the prompt.

4 Why Text Does Not Need to Be Exact

The final architecture treats text as an auxiliary semantic cue, not as the central audio representation. This is a deliberate difference from a cascaded vision-to-text-to-audio system.

In a cascade, recognition errors in the text stage directly become content errors in the audio stage. In our model, lip motion and audio-latent prompt conditions remain visible to the reconstructor. The text hidden state provides coarse semantic guidance and helps stabilize alignment, but it does not have to carry phonetics, speaker timbre, timing, and acoustic detail by itself.

The strongest evidence is the ground-truth text versus AVSR text evaluation:

Text condition	Corr	MSE	MAE
Ground-truth text + word timestamps	0.58184180	0.66763106	0.60181897
Auto-AVSR text	0.57833471	0.67162028	0.60371083

Table 3: Full LRS3 validation metrics on 1,000 clips. Replacing ground-truth text with Auto-AVSR text drops correlation by only 0.00351.

This does not mean text is useless. It means transcript accuracy is not the limiting factor in this architecture. Even when the text is imperfect, generated audio can remain perceptually acceptable if the lip sequence and audio-latent conditions are matched. Therefore, the project focus should remain on visual speech representation, audio latent modeling, and timbre control rather than on building a high-accuracy intermediate transcript.

The practical implication is important for system design. The text condition can be noisy, and even substantial recognition errors should not be expected to damage the audio in the same way they would damage a text-first cascade, because the audio reconstructor still receives lip motion and audio-latent/timbre conditions. We did not run a separate controlled word-error-rate sweep, so this report does not claim a fixed WER threshold. The measured result is more specific: replacing ground-truth text with the available Auto-AVSR text barely changes the final latent metric.

5 Experimental Results

5.1 Final Checkpoint

- **Config:** `configs/fm_avsr_lipavsr_59144_timbre3s_audioprompt38_pool_promptstats005_residual_samplecorr02_from1000_recon_textjson_wordts.yaml`
- **Checkpoint:** `runs/fm_avsr/lipavsr_59144_timbre3s_audioprompt38_pool_promptstats005_residual_samplecorr02_from1000_recon_textjson_wordts_v1/step_001500.pt`
- **Residual base:** `runs/fm_avsr/lipavsr_59144_timbre3s_audioprompt38_pool_from4000_recon_textjson_wordts_v1/step_005000.pt`

5.2 Scale and Timbre Ablations

The 10k to 30k scale-up improved matched validation correlation by about 0.0236. Timbre conditioning gave a larger change: the 30k no-timbre baseline at 0.5330 moved to 0.5633 with mean/std timbre, and the prompt/residual route reached 0.5726. The final 59k residual prompt-statistics checkpoint reached 0.5818.

The shifted-condition negative control for the 30k audio-prompt model dropped to 0.00799 correlation, confirming that the model uses the matched conditions rather than predicting an average latent.

Setup	Corr	MSE	MAE
10k visual-prior baseline	0.50725122	0.77212131	0.65607136
30k visual-prior baseline	0.53081070	0.74379597	0.64330638
30k no-timbre baseline	0.53298334	0.72177195	0.63384026
30k mean/std timbre	0.56328889	0.68858632	0.61404544
30k prompt tokens + pooled prompt + residual	0.57255184	0.67793650	0.60844724
Final 59k prompt-stats residual	0.58184180	0.66763106	0.60181897

Table 4: Main metric progression. Data scale improves the visual-prior model, and timbre/audio prompt conditioning gives a larger gain than text-source changes.

5.3 Raw and Silent Video Inference

External raw-video inference uses Auto-AVSR text and uniform text alignment because no ground-truth transcript or word timestamps exist. The final raw-video path is therefore the same conceptual architecture as training, but with the text source switched from `text_json` to `lipavsr`.

For silent video, the video supplies lip and face conditions while an optional reference audio supplies A and q . If no reference audio is provided, the audio prompt and timbre vector are zero. The current demo export drops the first 3.04 seconds because the model can copy reference-prompt audio into the beginning of the generated waveform. This is a temporary export-time hack and is listed as a model issue to fix.

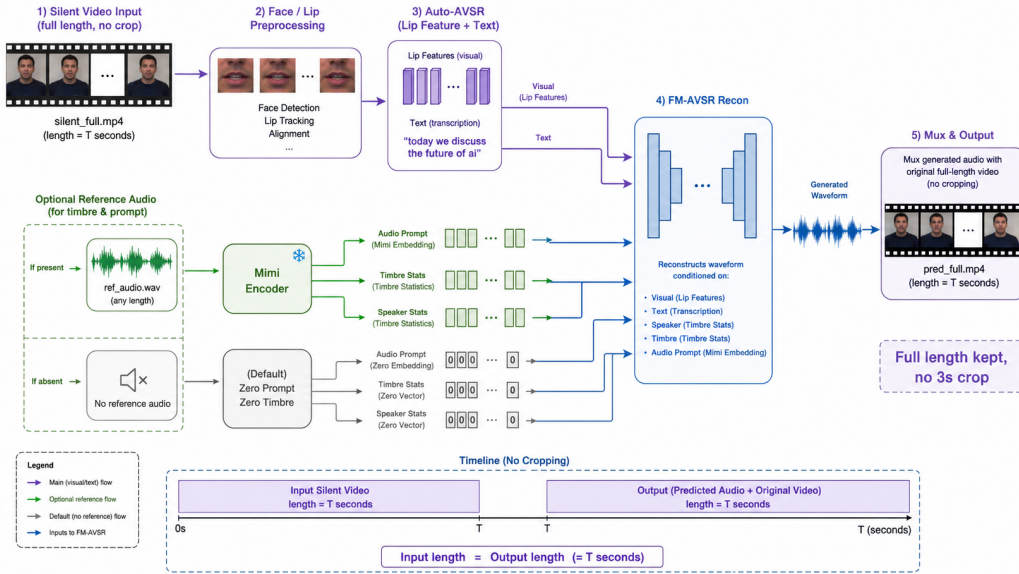


Figure 3: Silent-video inference with optional reference audio. This is an application of the same audio-prompt/timbre conditioning mechanism used in training. Current listening exports skip the first 3.04 seconds to avoid reference-prompt leakage at the start.

6 Reproducibility Interfaces

The primary command-line entry point is:

```
python scripts/run_raw_video_avsr_recon_pipeline.py \
  --input data/trump.mov --exp trump_cleanup_verify --force
```

For silent video with reference audio:

```
python scripts/run_raw_video_avsr_recon_pipeline.py \
```

```
--input silent.mp4 --ref_audio ref.wav \  
--silent_input --exp demo --force
```

The Gradio interface at `scripts/gradio_avsr_gui.py` exposes the same backend with video upload, optional reference audio, and silent-input controls.

These interfaces are not counted as research contributions; they are included so the result can be reproduced and demonstrated.

7 Limitations

- The system is offline. Earlier timing showed that face alignment and lip cropping dominate runtime; the FM head and Mimi decoder are not the primary bottlenecks.
- The model is deterministic. It improves stability, but it can average speaker-specific acoustic detail, which explains some mixed-voice artifacts.
- Timbre control is based on prompt latents and prompt statistics. It is useful, but it is not equivalent to a high-fidelity speaker cloning model.
- The current silent-reference export skips the first 3.04 seconds because the model can copy prompt/reference audio at the beginning. A proper fix should prevent prompt leakage inside the model rather than relying on export-time cropping.
- External videos remain harder than LRS3 because of domain shift, crop quality, missing word timestamps, and imperfect Auto-AVSR text.
- Latent correlation is a useful paired-condition metric, but it is not a complete subjective audio-quality score.

8 Conclusion

The final FM-AVSR result is best understood as audio-centric visual speech reconstruction. The model does not rely on a perfect intermediate transcript. It uses lip evidence, weak text, speaker information, and audio-prompt timbre conditions to reconstruct Mimi latents directly. The strongest empirical evidence is that replacing ground-truth text with Auto-AVSR text only reduces validation correlation by 0.0035, while timbre/audio-prompt conditioning and residual endpoint reconstruction produce the major gains. This supports the chosen architecture over a strict vision-to-text-to-audio cascade and identifies future improvement targets: better lip/audio representations, stronger timbre control, and higher-fidelity latent decoding.